Data Structure and Algorithm II Lab

Marks-30

Problem 1: Second Minimum (Set-1)

Given an array containing N distinct integers, write a program using the **Divide and Conquer** technique to find the second minimum element of the array.

Input Format

The first line contains an integer N, representing the size of the array. The second line contains N space-separated distinct integers representing the elements of the array.

Input:

6

8 3 5 1 9 6

Output:

3

Problem 2: Three-Way Merge Sort (Set-2)

Given an integer array arr of size N, sort the array using the **Divide and Conquer** technique.

Unlike the conventional Merge Sort, at every recursive step you must divide the current array into **three parts**, as evenly as possible. Recursively sort the three parts and then merge the three sorted parts to obtain the final sorted array.

Input Format

The first line contains an integer N, representing the size of the array.

The second line contains N space-separated integers representing the elements of the array.

Input:

8
9 4 7 1 3 8 2 6

Output:

1 2 3 4 6 7 8 9

Problem 3: Cargo Loading with Processing Cost (Set-1)

A delivery company has a truck with a maximum carrying capacity of W kilograms. There are N different cargo items. Each item has a given weight and a profit obtained by delivering the complete item.

The company is allowed to load either a complete item or only a portion of an item. If only a fraction of an item is loaded, the profit obtained from that item is proportional to the fraction loaded.

However, splitting an item requires additional handling and packaging. Therefore, whenever an item is **partially loaded**, a fixed processing cost C is deducted from the profit obtained from that item. No processing cost is charged when an item is loaded completely.

Determine the maximum profit that can be obtained without exceeding the capacity of the truck.

Input Format

The first line contains three values: N W C

where N is the number of cargo items, W is the maximum capacity of the truck, and C is the processing cost for partially loading an item.

The second line contains N space-separated integers representing the weights of the items. The third line contains N space-separated integers representing the profits of the complete items.

Input:

3 50 5
10 20 30
60 100 120

Output:

235

Problem 4: Task Scheduling with a Lunch Break (Set-2)

A person has N tasks planned during a working day. Each task has a starting time and an ending time. The person can perform only **one task at a time**, so two selected tasks cannot overlap.

Additionally, the person has a mandatory lunch break. The starting and ending times of the lunch break are also given as input. No selected task may overlap with the lunch-break interval.

Find the maximum number of tasks that the person can complete during the day.

Input Format

⇒ The first line contains an integer N, representing the number of tasks.

⇒ The second line contains two integers: **LStart LEnd.** representing the starting and ending times of the lunch break in 24-hour HHMM format.

⇒ The next N lines each contain two integers: **Start End.** representing the starting and ending times of one task.

Input:

6
1200 1300
900 1000
930 1030
1030 1130
1130 1230
1300 1400
1400 1500

Output:
4

Problem 5: Flexible Coin Change (set 1 and 2)

You are given N different coin types and a target change amount X.

Normally, the exact amount X should be returned. However, in this system a difference of 1 unit is acceptable. Therefore, you are allowed to return any one of the following amounts:

X - 1, X, X + 1

Using the available coin types, find the minimum number of coins required to produce any one of these three amounts.

Input Format

The first line contains two integers:

N X

where N is the number of coin denominations and X is the requested change. The second line contains N space-separated integers representing the coin denominations.

Hint

If the requested change is 12, you should consider the minimum number of coins required to make: 11, 12, and 13. Then take the minimum among these three possibilities.

Sample 1

Input:

2 12
3 7

Output:

3

Here, 12 requires four coins (3 + 3 + 3 + 3), while 13 can be formed using three coins (7 + 3 + 3). Therefore, the answer is 3.